TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Acceleration Due to Gravity by a Compound Pendulum

The bar pendulum — period against suspension distance, the equivalent simple pendulum found from the graph, and g computed from L and T.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 1 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to determine g with a compound (bar) pendulum. A rigid body swinging about a horizontal axis at distance l from its centre of gravity G has period

$$T = 2\pi \sqrt{((k^2 + l^2) / (g\, l))}$$

where k is the radius of gyration about G. The same period belongs to a simple pendulum of length $L = l + k^2/l$ — the **equivalent simple pendulum**, whose bob would sit at the **centre of oscillation**. Because $l + k^2/l$ appears symmetrically, two suspension distances on opposite sides of G give the *same* period with the same L — and that symmetry is the whole method: **L is simply the distance between the two knife edges that give equal periods**, and then

$$g = 4\pi^2 L / T^2$$

with no need to know k or the mass distribution at all.

2. Apparatus and procedure

- **Apparatus:** a uniform metal bar (~1 m) pierced with a row of holes, two knife-edge supports, stopwatch, metre rule.
- Mark the centre of the bar (for a uniform bar, G is at its midpoint). Suspend from a hole near one end; set small oscillations (amplitude under about 5°, in one plane); time 20 oscillations; compute T.
- Repeat hole by hole, moving the suspension point along the bar through G to the far end, recording T at each distance from one end.
- Plot T (y-axis) against distance along the bar (x-axis). The curve is symmetric about G with two minima. Draw a horizontal line at a chosen T cutting the curve at four points A, B (left of G) and C,

D (right of G): the pairs (A, D) and (B, C) are the two equal-period knife-edge pairs. L = distance AD = distance BC; average them.

- Compute $g = 4\pi^2 L/T^2$ for the chosen T, and repeat for a second horizontal line as a check.

3. Worked calculation (model data)

RESULT

From the graph, for T = 1.84 s the equal-period knife edges are 84.0 cm apart (average of the two pairs): L = 0.840 m.

$g = 4\pi^2 L/T^2 = (39.478 \times 0.840) / (1.84)^2 = 33.16 / 3.386 =$ **$9.79\ \text{m s}^{-2}$**.

The expected value at Dhaka's latitude is about 9.79–9.80 m s^{-2}, so the model result is right where a careful run lands. The minimum period on the curve, where l = k, is the sharpest single check of the timing.

4. Precautions

- Keep the amplitude small (the period formula assumes small angles) and the swing in one vertical plane — elliptical swinging falsifies T.
- The knife edges must be truly horizontal and sharp; a rocking or slipping edge adds its own motion.
- Time many oscillations (20–30) and start counting at a zero crossing, "zero" called at the first pass; reaction time then divides by the count.
- Use two different horizontal lines on the graph: consistent L values from both are the experiment's internal proof.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Define the centre of oscillation, and state why the distance between two equal-period knife edges equals the equivalent length L.
2. From a graph, equal-period edges are 78.0 cm apart at T = 1.775 s. Compute g.
3. Why does the T-curve have two minima, and at what suspension distance is T smallest?
4. Why is the compound pendulum preferred to a simple pendulum for measuring g in a teaching laboratory?
5. A student times 20 oscillations but counts the first pass as "one". What error enters T, and in which direction does g move?
6. Why must the amplitude be kept small?

Answer key

1. The centre of oscillation is the point on the line through G, at distance $L = l + k^2/l$ from the axis, where the whole mass could be concentrated without changing the period. If two axes at distances l_1 and l_2 (on opposite sides of G, $l_1 \neq l_2$) give the same T, then $l_1 + k^2/l_1 = l_2 + k^2/l_2$ forces $l_1 l_2 = k^2$, so $L = l_1 + l_2$ — the separation of the two knife edges. Measuring that separation therefore measures L directly.
2. $g = 4\pi^2 \times 0.780/(1.775)^2 = 30.79/3.151 = 9.77$ m s^{-2}.
3. Because T depends on $(k^2 + l^2)/l$, which behaves the same whether the axis is at l or at k^2/l — one minimum on each side of G. T is smallest when the derivative vanishes, at $l = k$, where $L = 2k$.
4. A simple pendulum's length must be measured to the centre of a bob whose support stretches and whose "point mass" is an idealisation; its period also depends on the thread. The bar pendulum needs only a distance between knife edges and a time — both measurable precisely — and its theory absorbs the mass distribution into the graph.
5. The 20 timed intervals are really 19, so the computed T is too small by a factor 19/20; since $g \propto 1/T^2$, g comes out too large — by about 10%, a huge and classic blunder.
6. The exact period grows with amplitude (the restoring torque is mg sinθ, and the linearised formula replaces sinθ by θ). Small amplitudes keep that correction negligible, so $g = 4\pi^2 L/T^2$ applies as written.

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